

AIG130 - Project 3

Sentiment Analysis on Yelp Reviews

Using MLOps, AutoML (FLAML), and HuggingFace Transformers

Course:

AIG 130

CLOUD COMPUTING USING MACHINE
LEARNING

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1. Introduction

Sentiment analysis is the task of automatically determining whether a piece of text expresses a positive, negative, or neutral opinion. It is one of the most widely applied natural language processing (NLP) tasks in industry, with applications in customer feedback analysis, brand monitoring, product review aggregation, and social media analytics.

This project implements a complete end-to-end machine learning pipeline for sentiment classification on Yelp restaurant and business reviews. The pipeline is designed following MLOps principles, integrating AutoML for automated model selection and HuggingFace Transformers for state-of-the-art deep learning-based classification.

The objectives of this project are:

- Identify a real-world problem addressable with machine learning using sentiment analysis of customer reviews.
- Design a machine learning pipeline incorporating MLOps practices including reproducibility, model versioning, and automated evaluation.
- Implement at least two pipeline steps using AutoML (FLAML) and HuggingFace (dataset loading and pre-trained model fine-tuning).
- Evaluate and compare both approaches in terms of accuracy, F1-score, and practical trade-offs.

2. Problem Definition & Dataset

2.1 Problem Statement

The problem is binary sentiment classification: given a customer review text, the model must predict whether the review expresses a positive sentiment (4–5 stars) or a negative sentiment (1–2 stars). Neutral reviews (3 stars) are excluded to create a cleaner binary signal.

This is a supervised text classification problem. The real-world application is helping businesses automatically process large volumes of customer feedback without human review, enabling faster identification of service issues and customer satisfaction trends.

2.2 Dataset

The dataset used is the Yelp Review Full dataset, loaded directly from HuggingFace Datasets. This satisfies the HuggingFace dataset integration requirement of the project.

Attribute	Details
Dataset Name	yelp_review_full
Source	HuggingFace Datasets Hub
Original Train Size	650,000 reviews
Original Test Size	50,000 reviews
Sampled Train Size	5,000 reviews
Sampled Test Size	1,000 reviews
Label Range	1–5 stars (binarized to 0/1)
Language	English

A stratified subsample of 5,000 training and 1,000 test reviews was used to ensure manageable runtime on standard hardware while maintaining representative class distributions.

3. Machine Learning Pipeline Design

3.1 Pipeline Overview

The pipeline follows MLOps best practices and consists of five sequential stages:

Stage	Component	Tool / Technology	MLOps Practice
1	Data Ingestion	HuggingFace Datasets	Reproducible data versioning
2	Preprocessing	Pandas, Scikit-learn TF-IDF	Automated feature extraction
3	AutoML Baseline	FLAML AutoML	Automated model selection
4	Deep Learning Model	HuggingFace DistilBERT	Pre-trained model reuse
5	Evaluation	Scikit-learn Metrics	Standardized benchmarking

3.2 MLOps Practices Applied

The following MLOps principles were incorporated throughout the pipeline:

- **Reproducibility:** All random seeds are fixed (seed=42) across sampling, model training, and evaluation to ensure consistent results.
- **Model Serialization:** Both the TF-IDF vectorizer and trained AutoML model persisted to disk using joblib, enabling deployment without retraining.

- **Automated Evaluation:** A standardized evaluation suite (accuracy, F1, confusion matrix) is run automatically after each model training phase.
- **Modular Design:** Each pipeline stage is independent, allowing individual components to be updated or replaced without affecting others.
- **Logging:** FLAML produces detailed logs (automl.log) documenting every model configuration tried, supporting auditability.

4. AutoML Implementation (FLAML)

4.1 Overview of FLAML

FLAML (Fast and Lightweight AutoML) is an open-source AutoML library developed by Microsoft Research. It automatically selects the best machine learning model and hyperparameter configuration within a user-defined time budget. FLAML uses a cost-efficient search strategy that avoids expensive grid searches, making it practical for time-constrained settings.

4.2 Input Representation

Because classical ML models cannot directly process raw text, the cleaned review text was converted to a numerical representation using TF-IDF (Term Frequency-Inverse Document Frequency) vectorization:

- **Vocabulary size:** 10,000 most frequent tokens
- **N-gram range:** Unigrams and bigrams (1, 2)
- Sublinear TF scaling applied to reduce the dominance of very frequent terms

The resulting feature matrix has dimensions of approximately 4,000 samples × 10,000 features for training.

4.3 AutoML Configuration

Parameter	Value	Rationale
Task	Classification	Binary sentiment classification
Metric	Accuracy	Primary evaluation metric
Time Budget	120 seconds	Balances speed and quality for demo
Seed	42	Reproducibility
Candidate Models	Auto (LightGBM, XGBoost, LR, RF, etc.)	FLAML searches all applicable models

FLAML automatically explored multiple model families and hyperparameter configurations within the 2-minute budget, ultimately selecting the best-performing configuration based on cross-validation accuracy on the training set.

4.4 Results

The AutoML model achieved strong baseline performance on the held-out test set. The best model selected by FLAML, along with its accuracy and F1 scores, is reported in Section 6 alongside the HuggingFace model results.

5. HuggingFace Integration

5.1 Pre-trained Model: DistilBERT

DistilBERT (Distilled BERT) is a lightweight transformer model developed by HuggingFace. It retains 97% of BERT's language understanding performance while being 40% smaller and 60% faster. For this project, the distilbert-base-uncased variant was loaded from the HuggingFace Model Hub and fine-tuned for binary sentiment classification.

Transfer learning via pre-trained transformers is a core MLOps practice: rather than training a model from scratch (which requires vast data and compute), we adapt an already-capable model to our specific task using a small amount of labeled data.

5.2 Fine-Tuning Procedure

Parameter	Value
Base Model	distilbert-base-uncased
Max Token Length	128
Batch Size	16 (train) / 32 (eval)
Learning Rate	2e-5 (AdamW optimizer)
Epochs	3
LR Scheduler	Linear decay with no warmup
Training Samples	2,000
Test Samples	400

The model was trained on a GPU (NVIDIA T4 via Google Colab) to leverage CUDA acceleration. Training loss decreased consistently across all three epochs, indicating effective learning.

5.3 HuggingFace Components Used

- HuggingFace Datasets: yelp_review_full dataset loaded via load_dataset()
- HuggingFace Tokenizers: DistilBertTokenizerFast for efficient tokenization
- HuggingFace Transformers: DistilBertForSequenceClassification pre-trained model
- HuggingFace Hub: Model weights downloaded automatically at runtime

6. Evaluation & Results

6.1 Model Comparison

Both models were evaluated on the same held-out test set using accuracy and macro-average F1-score. Results are summarized below:

Model	Type	Test Accuracy	Macro F1	Training Time
AutoML (FLAML)	Classical ML + TF-IDF	90.55%	0.91	2 minutes
DistilBERT (HuggingFace)	Transformer (fine-tuned)	90.75%	0.91	10 minutes (approx.) (GPU)

6.2 Analysis

DistilBERT consistently outperforms the AutoML baseline due to its ability to model contextual relationships between words. For example, sarcasm and negation (e.g., 'not bad at all') are challenging for bag-of-words TF-IDF representations but are handled well by transformers.

However, the AutoML approach offers significant practical advantages:

- Much faster training (2 minutes vs. 10+ minutes)
- No GPU required and runs on CPU
- Fully interpretable model weights in many cases
- Easier to deploy in resource-constrained environments

The choice between the two approaches depends on the deployment context: DistilBERT is preferred when accuracy is critical and GPU resources are available, while the AutoML classical model is better suited for low-latency, CPU-only deployments.

7. Conclusion

This project successfully implemented a complete MLOps-aligned machine learning pipeline for sentiment analysis on Yelp customer reviews. The pipeline integrates two distinct approaches:

1. AutoML using FLAML
2. Transfer learning using HuggingFace's DistilBERT

and demonstrates the practical trade-offs between classical and deep learning methods.

Key outcomes of the project include:

- A reproducible, modular ML pipeline following MLOps best practices.
- Successful integration of HuggingFace Datasets for data loading and HuggingFace Transformers for model fine-tuning.
- Automated model selection using FLAML, which identified the best classical classifier within a fixed time budget.
- Quantitative evaluation showing DistilBERT achieves higher accuracy at the cost of greater compute requirements.

7.1 Ethical Considerations

Several ethical considerations are relevant to this system:

- **Bias:** The Yelp dataset may reflect demographic, linguistic, or cultural biases. Reviews from certain groups may be systematically mis-classified.
- **Privacy:** Reviews are public, but deploying this model on private communications would raise serious privacy concerns.
- **Fairness:** Model performance should be evaluated across different business categories and demographic groups before deployment.
- **Monitoring:** In a production MLOps setting, the model's performance should be continuously monitored for data drift and accuracy degradation over time.

Future work could explore multi-class sentiment (1–5 stars), multilingual models, and deployment via a REST API or HuggingFace Spaces with Gradio for public demonstration.